

Disk, Volume, and Quota Management

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Contents

Step 2: Create a Spanned Volume	5
Step 3: Create a Striped Volume	6
Step 4: Create a Mirrored Volume	7
Step 4: Create a RAID-5 Volume	8
Part C: Department Folder Structure and Sharing	9
Step 1: Create Department Folders	9
Step 2: Share Each Folder	10
Step 3: Verify in Active Directory	11
Part D: Quota Management using FSRM	12
Step 1: Install File Server Resource Manager	12
Step 2: Open FSRM	12
Step 4: Verify Quotas	13
Step 6: Test Quota Enforcement	14
Step 6: Create a Reusable Quota Template	Error! Bookmark not defined.

Part A: Disk Initialization and Conversion

Four virtual disks were initialized with the GPT partition style and converted to dynamic status.

The screenshot displays two windows from a Windows Server environment. The left window is the 'Server Manager' console for 'Local Server', showing system properties and event logs. The right window is 'Disk Management', showing the disk configuration.

Server Manager - Local Server Properties:

- Computer name: ServerDC
- Domain: contoso.com
- Microsoft Defender Firewall: Public: On
- Remote management: Enabled
- Remote Desktop: Disabled
- NIC Teaming: Disabled
- Ethernet0: 10.1.1.100
- Operating system version: Microsoft Windows Server 2022 Datacenter
- Hardware information: VMware, Inc. VMware20.1

Server Manager - Events:

Server Name	ID	Severity	Source
SERVERDC	10016	Warning	Microsoft-Windows-DistributedCOM
SERVERDC	1046	Error	Microsoft-Windows-DHCP-Server
SERVERDC	1059	Error	Microsoft-Windows-DHCP-Server
SERVERDC	1059	Error	Microsoft-Windows-DHCP-Server
SERVERDC	12	Warning	Microsoft-Windows-Time-Service
SERVERDC	12317	Warning	SRMSVC

Disk Management:

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...	59.68 GB	45.97 GB	77 %
(Disk 4 partition 1)	Simple	Basic		Healthy (E...	200 MB	200 MB	100 %

Disk Details:

- Disk 0: Dynamic, 1.98 GB, Online, Unallocated
- Disk 1: Dynamic, 1.98 GB, Online, Unallocated
- Disk 2: Dynamic, 1.98 GB, Online, Unallocated
- Disk 3: Dynamic, 1.98 GB, Online, Unallocated
- Disk 4: Basic, 59.68 GB, Online, 200 MB (C:), 59.68 GB NTFS, Healthy (Boot Data File, Crash Dump, Basic Data Partition), Unallocated, Primary partition

Part B: Creating Volumes

Step 1: Create a Simple Volume

Uses the available unallocated space on the first dynamic disk to create a new simple volume with the drive letter S, the NTFS format, and labeled “SIMPLE”.

The screenshot displays two windows from a Windows Server environment. The left window is 'Server Manager' for 'Local Server', showing the 'PROPERTIES' tab for 'ServerDC'. It lists system information such as Computer name (ServerDC), Domain (contoso.com), and Operating system version (Microsoft Windows Server 2022 Datacenter). Below this is the 'EVENTS' section, showing a list of events with columns for Server Name, ID, Severity, and Source. The right window is 'Disk Management', showing a list of volumes and a detailed view of 'Disk 0'. The volume list shows '(C:)', '(Disk 4 partition 1)', and 'SIMPLE (S:)'. The detailed view for 'Disk 0' shows a 'SIMPLE (S:)' volume of 200 MB NTFS, which is healthy and has 1.79 GB of unallocated space. The bottom of the screen shows the Windows taskbar with the search bar and system clock indicating 12:37 AM on 11/24/2025.

Step 2: Create a Spanned Volume

Uses the available unallocated space on the first dynamic disk to create a new spanned volume with the drive letter P, the NTFS format, and labeled “SPANNED”.

The screenshot displays two windows from a Windows Server environment. The left window is 'Server Manager' for 'Local Server', showing system properties for 'ServerDC' (contoso.com) and a list of events. The right window is 'Disk Management', showing a table of volumes and a graphical representation of disks.

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...)	59.68 GB	45.97 GB	77 %
(Disk 4 partition 1)	Simple	Basic	NTFS	Healthy (E...)	200 MB	200 MB	100 %
SIMPLE (S)	Simple	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
SPANNED (P)	Spanned	Dynamic	NTFS	Healthy	400 MB	384 MB	96 %

Disk	Type	Capacity	Layout	Volume	File System	Status
Disk 0	Dynamic	1.98 GB	Online	SIMPLE (S)	200 MB NTFS	Healthy
				SPANNED (P)	200 MB NTFS	Healthy
					1.59 GB	Unallocated
Disk 1	Dynamic	1.98 GB	Online	SPANNED (P)	200 MB NTFS	Healthy
					1.79 GB	Unallocated
Disk 2	Dynamic	1.98 GB	Online			1.98 GB Unallocated
Disk 3	Dynamic	1.98 GB	Online			1.98 GB Unallocated
Disk 4	Basic	59.68 GB	Online	(C:)	59.68 GB NTFS	Healthy (Bios, Basic Data Partition)

The Disk Management window also includes a legend at the bottom: Unallocated (black), Primary partition (blue), Simple volume (green), and Spanned volume (purple).

Step 3: Create a Striped Volume

Uses the available unallocated space on the first dynamic disk to create a new striped volume with the drive letter R, the NTFS format, and labeled “STRIPE”.

The screenshot displays two windows from a Windows Server environment. The left window is 'Server Manager' for 'Local Server', showing system properties and events. The right window is 'Disk Management', showing the disk layout and the creation of a striped volume.

Server Manager - Local Server Properties:

- Computer name: ServerDC
- Domain: contoso.com
- Microsoft Defender Firewall: Public On
- Remote management: Enabled
- NIC Teaming: Disabled
- Ethernet0: 10.1.1.100
- Operating system version: Microsoft Windows Server 2022 Datacenter
- Hardware information: VMware, Inc. VMware20.1

Events: All events | 824 total. Filter: [X] [Y] [Z].

Server Name	ID	Severity	Source	Log
SERVERDC	2	Warning	volmgr	System
SERVERDC	2	Warning	volmgr	System
SERVERDC	2	Warning	volmgr	System
SERVERDC	2	Warning	volmgr	System
SERVERDC	2	Warning	volmgr	System
SERVERDC	2	Warning	volmgr	System

Disk Management:

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...	59.68 GB	45.97 GB	77 %
(Disk 4 partition 1)	Simple	Basic	NTFS	Healthy (E...	200 MB	200 MB	100 %
SIMPLE (S)	Simple	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
SPANNED (P)	Spanned	Dynamic	NTFS	Healthy	400 MB	384 MB	96 %
STRIPE (R)	Striped	Dynamic	NTFS	Healthy	2.34 GB	2.33 GB	99 %

Disk Layout:

- Disk 0:** Dynamic, 1.98 GB, Online. Contains SIMPLE (S:) 200 MB NTFS, SPANNED (P) 200 MB NTFS, and STRIPE (R:) 600 MB NTFS. 1.01 GB Unallocated.
- Disk 1:** Dynamic, 1.98 GB, Online. Contains SPANNED (P) 200 MB NTFS and STRIPE (R) 600 MB NTFS. 1.20 GB Unallocated.
- Disk 2:** Dynamic, 1.98 GB, Online. Contains STRIPE (R) 600 MB NTFS. 1.40 GB Unallocated.
- Disk 3:** Dynamic, 1.98 GB, Online. Contains STRIPE (R) 600 MB NTFS. 1.40 GB Unallocated.
- Disk 4:** Basic, 59.68 GB, Online. Contains (C:) 59.68 GB NTFS. 200 MB Unallocated.

Legend: Unallocated (Blue), Primary partition (Green), Simple volume (Yellow), Spanned volume (Red), Striped volume (Purple).

Step 4: Create a Mirrored Volume

Uses the available unallocated space on the first dynamic disk to create a new mirrored volume with the drive letter M, the NTFS format, and labeled “MIRROR”.

The screenshot displays two windows from a Windows Server environment. The left window is 'Server Manager' for 'Local Server', showing system properties for 'ServerDC' (contoso.com) and a list of events. The right window is 'Disk Management', showing a table of volumes and a graphical representation of five disks with their partitions.

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...)	59.68 GB	45.97 GB	77 %
(Disk 4 partition 1)	Simple	Basic	NTFS	Healthy (E...)	200 MB	200 MB	100 %
MIRROR (M)	Mirror	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
SIMPLE (S)	Simple	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
SPANNED (P)	Spanned	Dynamic	NTFS	Healthy	400 MB	384 MB	96 %
STRIPE (R)	Striped	Dynamic	NTFS	Healthy	2.34 GB	2.33 GB	99 %

Disk	Layout	Type	File System	Status	Capacity	Free Space
Disk 0	SIMPLE (S:)	200 MB NTFS	Healthy	200 MB NTFS	Healthy	832 MB Unallocated
Disk 1	SPANNED (P:)	200 MB NTFS	Healthy	600 MB NTFS	Healthy	1,01 GB Unallocated
Disk 2	STRIPE (R:)	600 MB NTFS	Healthy	1,40 GB	Unallocated	
Disk 3	STRIPE (R:)	600 MB NTFS	Healthy	1,40 GB	Unallocated	
Disk 4	(C:)	59.68 GB NTFS	Healthy (B:)	200 MB (E:)	200 MB (S:)	400 MB (P:)

The Disk Management window also includes a legend at the bottom: Unallocated (black), Primary partition (blue), Simple volume (green), Spanned volume (purple), Striped volume (orange), and Mirrored volume (red).

Step 4: Create a RAID-5 Volume

Uses the available unallocated space on the first dynamic disk to create a new RAID-5 volume with the drive letter V, the NTFS format, and labeled “RAID5”.

The screenshot displays two windows from a Windows Server environment. The left window is 'Server Manager - Local Server', showing the 'PROPERTIES' for 'ServerDC' (contoso.com). It lists various services like Microsoft Defender Firewall, Remote Desktop, and Network Teaming. The right window is 'Disk Management', showing a list of volumes and a graphical representation of five disks (Disk 0 to Disk 4) with their partitions and RAID configurations.

Volume List:

Volume	Layout	Type	File System	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (B...)	59.68 GB	45.97 GB	77 %
(Disk 4 partition 1)	Simple	Basic	NTFS	Healthy (E...)	200 MB	200 MB	100 %
MIRROR (M)	Mirror	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
RAID5 (V)	RAID-5	Dynamic	NTFS	Healthy	1.17 GB	1.15 GB	99 %
SIMPLE (S)	Simple	Dynamic	NTFS	Healthy	200 MB	186 MB	93 %
SPANNED (P)	Spanned	Dynamic	NTFS	Healthy	400 MB	384 MB	96 %
STRIPE (R)	Striped	Dynamic	NTFS	Healthy	2.34 GB	2.33 GB	99 %

Disk Management Graphical View:

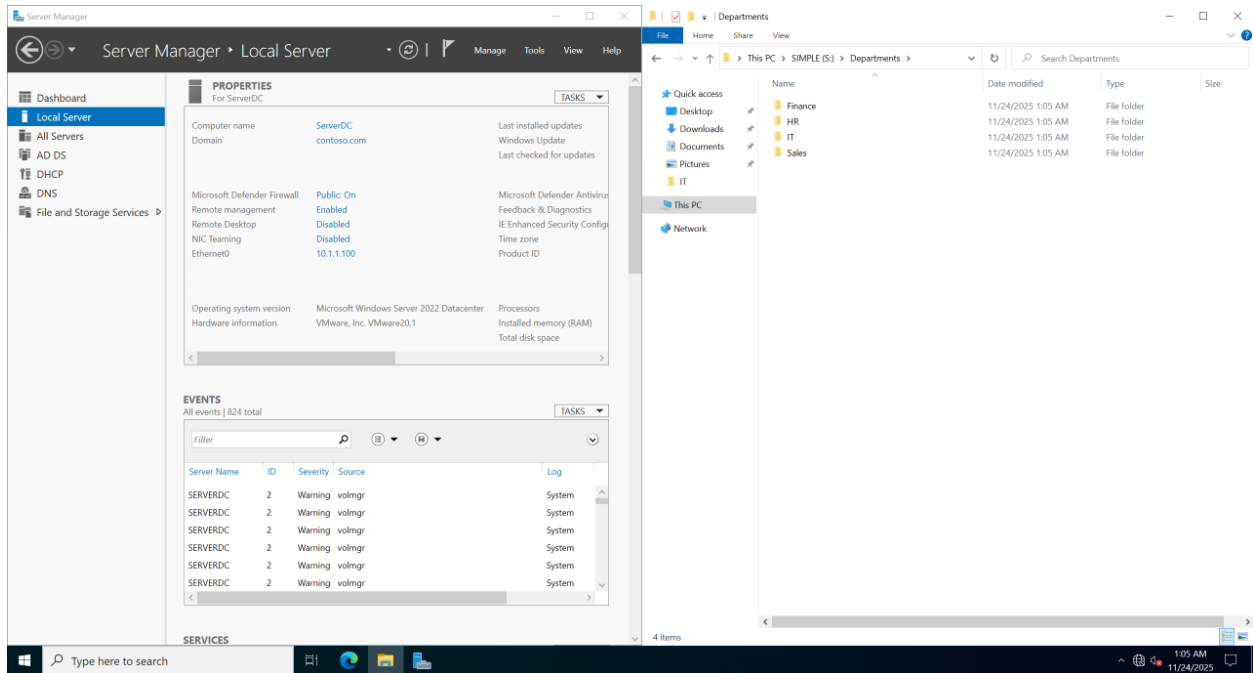
- Disk 0:** Dynamic, 1.98 GB, Online. Contains: SIMPLE (S) 200 MB NTFS, SPANNED (P) 200 MB NTFS, STRIPE (R) 600 MB NTFS, MIRROR (M) 200 MB NTFS, RAID5 (V) 600 MB NTFS, and 232 MB Unallocated.
- Disk 1:** Dynamic, 1.98 GB, Online. Contains: SPANNED (P) 200 MB NTFS, STRIPE (R) 600 MB NTFS, MIRROR (M) 200 MB NTFS, RAID5 (V) 600 MB NTFS, and 432 MB Unallocated.
- Disk 2:** Dynamic, 1.98 GB, Online. Contains: STRIPE (R) 600 MB NTFS and RAID5 (V) 600 MB NTFS, with 832 MB Unallocated.
- Disk 3:** Dynamic, 1.98 GB, Online. Contains: STRIPE (R) 600 MB NTFS and 1.40 GB Unallocated.
- Disk 4:** Basic, 59.68 GB, Online. Contains: (C:) 59.68 GB NTFS and 200 MB Unallocated.

A legend at the bottom of the Disk Management window identifies the colors for different volume types: Primary partition (blue), Simple volume (green), Spanned volume (yellow), Striped volume (orange), Mirrored volume (red), and RAID-5 volume (purple).

Part C: Department Folder Structure and Sharing

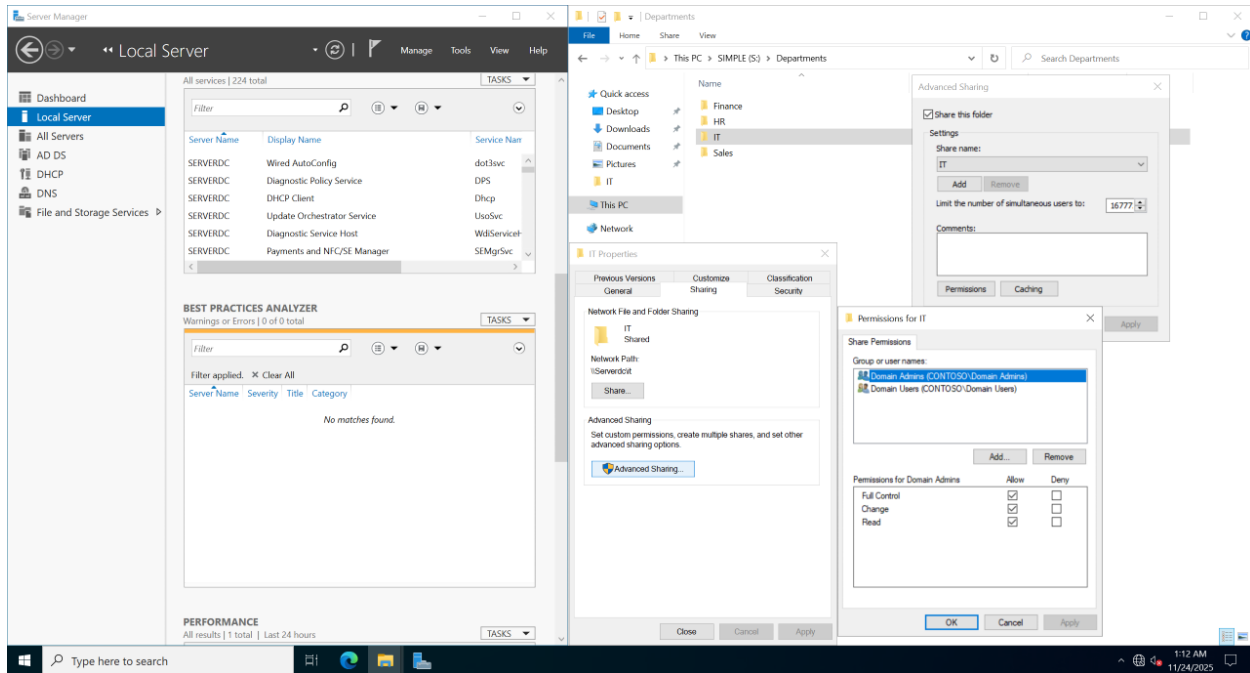
Step 1: Create Department Folders

Created a top-level departments folder on the data drive “S”, with four subfolders for IT, HR, Finance, and Sales.



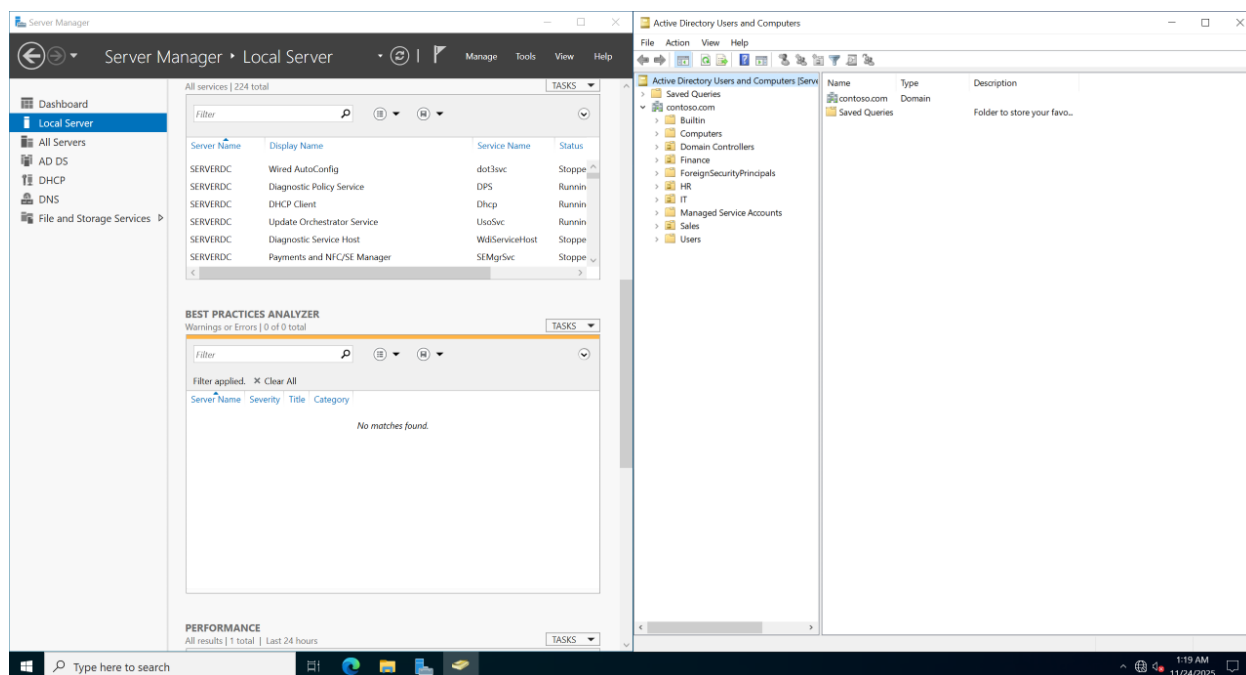
Step 2: Share Each Folder

Advanced sharing was used on each subfolder to grant full control to domain admins and change or read permissions to domain users.



Step 3: Verify in Active Directory

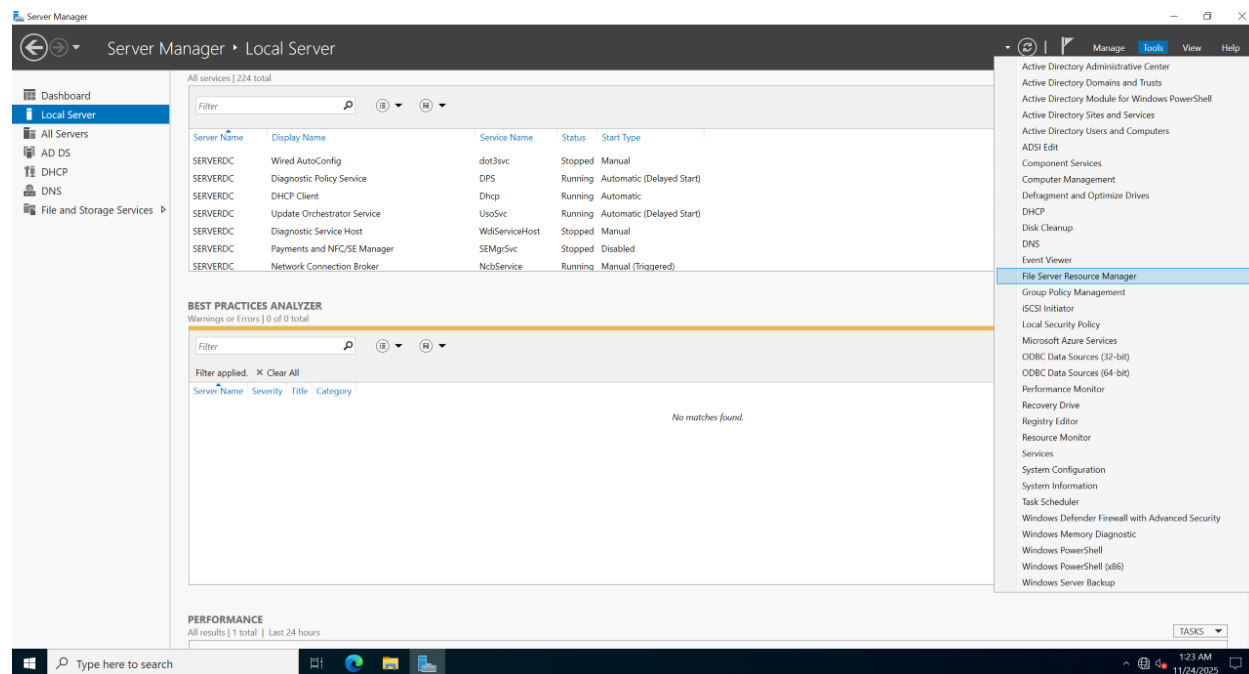
The corresponding IT, HR, Finance, and Sales organizational units were verified to exist in active directory users and computers.



Part D: Quota Management using FSRM

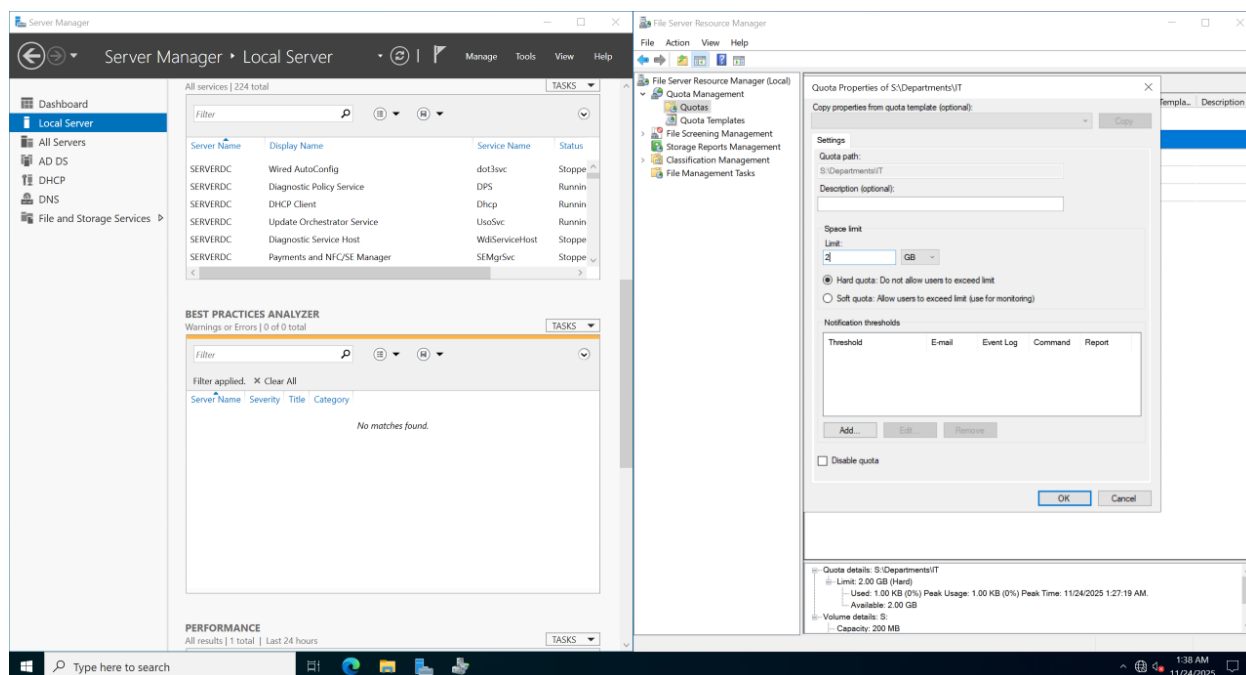
Step 1: Install File Server Resource Manager

File Server Resource Manager role service was installed using the server manager.

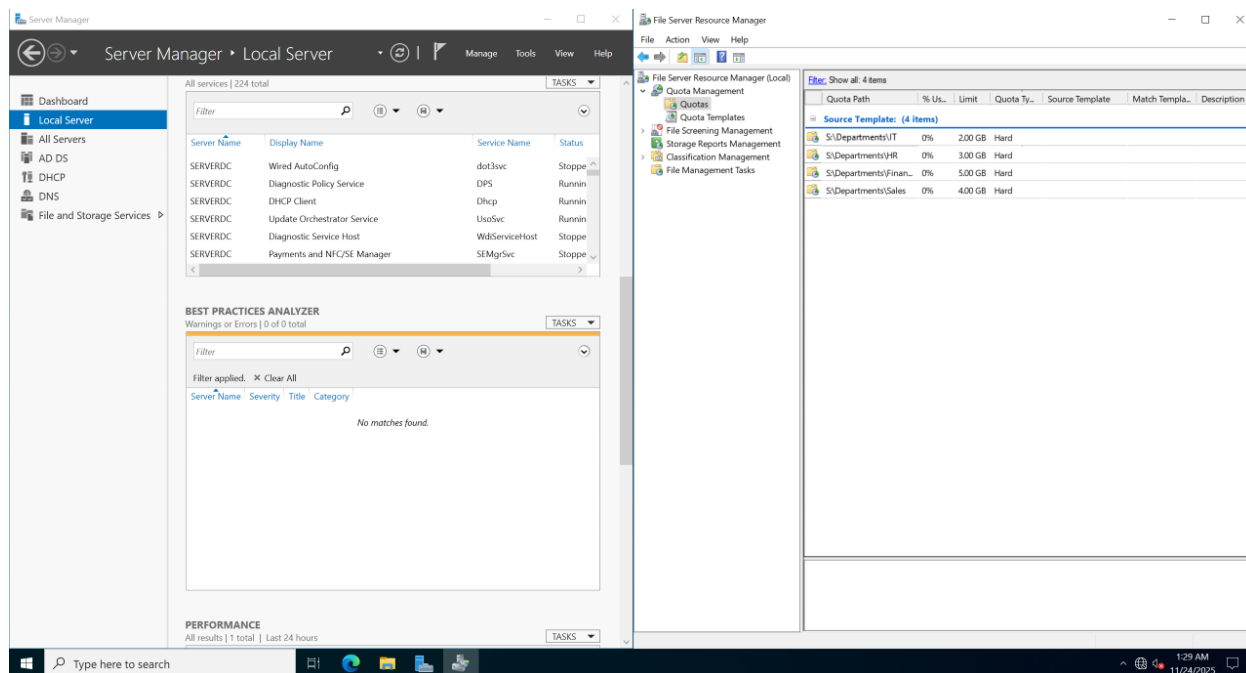


Step 3: Create Quotas for Each OU Folder

The “create quota” dialogue of the File Server Resource Manager console that was used to create a hard quotas for each department folder.



The File Server Resource Manager quota list that shows all department quotas.



Step 4: Verify Quotas

All four quotas were verified to appear correctly with viewing the File System Resource Manager.

The screenshot displays two windows from a Windows Server environment. The left window is the 'Server Manager' console, showing the 'Local Server' view with a list of services and a 'BEST PRACTICES ANALYZER' section. The right window is the 'File Server Resource Manager (Local)' console, showing the 'Quota Management' section. It lists four quotas for the S:\Departments folder:

Quota Path	% Us...	Limit	Quota Ty...	Source Template	Match Templa...	Description
S:\Departments\IT	0%	2.00 ...	Hard			
S:\Departments\HR	0%	3.00 GB	Hard			
S:\Departments\Finan...	0%	5.00 GB	Hard			
S:\Departments\Sales	0%	4.00 GB	Hard			

Below the table, the 'Quota details' for S:\Departments\IT are shown:

- Limit: 2.00 GB (Hard)
- Used: 1.00 KB (0%) Peak Usage: 1.00 KB (0%) Peak Time: 11/24/2025 1:27:19 AM
- Available: 2.00 GB
- Volume details: S
- Capacity: 200 MB

Folder properties of the IT department.

The screenshot shows the 'File Explorer' window displaying the 'Departments' folder. The 'IT' folder is selected. The 'IT Properties' dialog box is open, showing the 'General' tab. The properties are as follows:

Property	Value
Name	IT
Type	File folder
Location	S:\Departments
Size	0 bytes
Size on disk	0 bytes
Contains	0 Files, 0 Folders
Created	Monday, November 24, 2025, 1:05:04 AM
Attributes	☒ Read only (Only applies to files in folder) ☐ Hidden